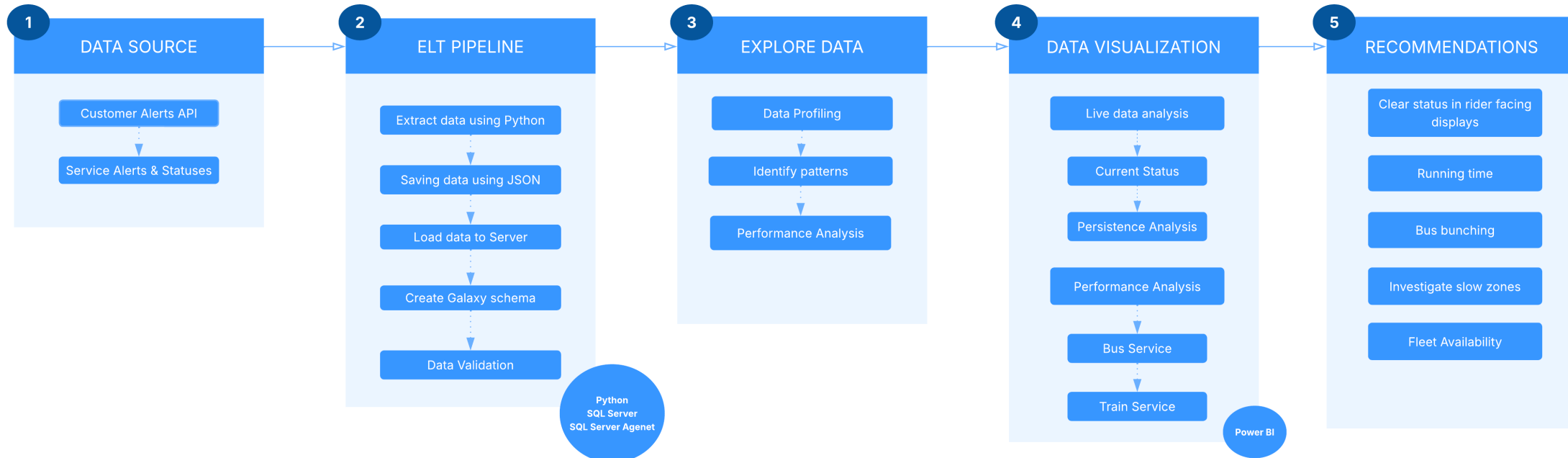
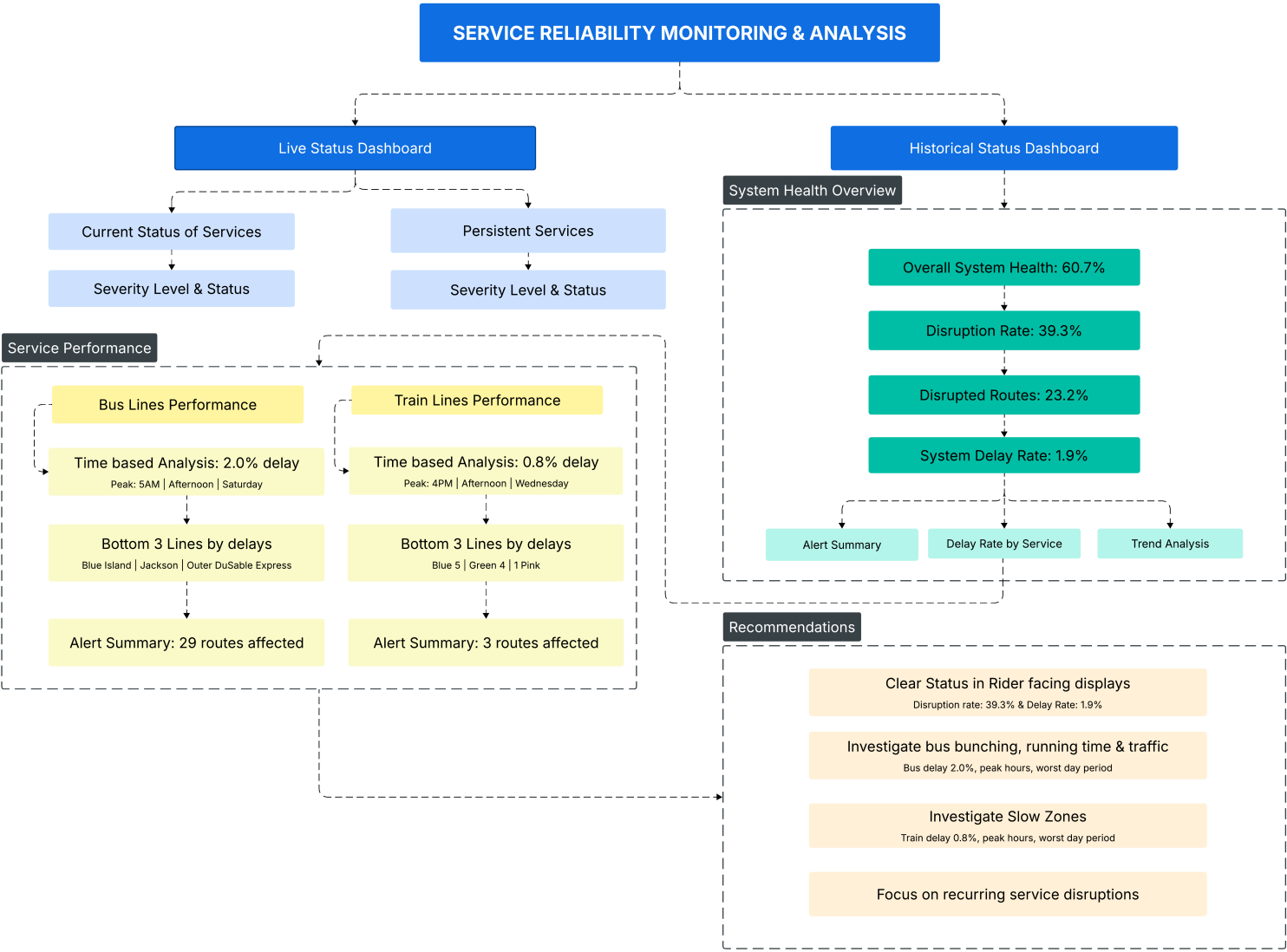


# PROJECT WORKFLOW

## Chicago Transit Authority (CTA) Performance Analysis



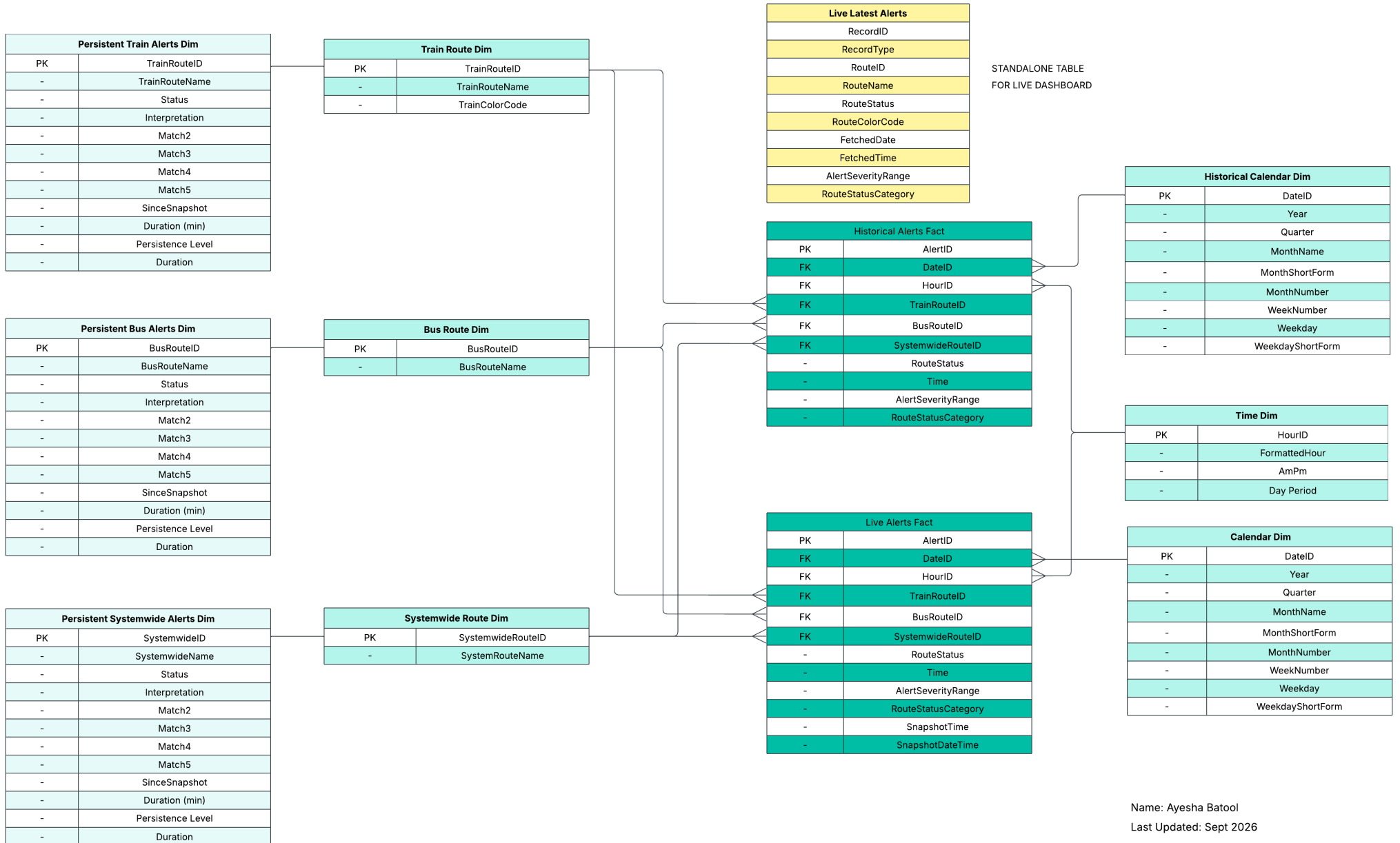
# ANALYTICAL WORKFLOW



# ENTITY RELATIONSHIP DIAGRAM (ERD)

Galaxy Schema

## Chicago Transit Authority (CTA) Service Reliability Analysis



# DATA DICTIONARY

| Column Name         | Data Type    | Description                                                                         | Key | Table Name             |
|---------------------|--------------|-------------------------------------------------------------------------------------|-----|------------------------|
| BusRouteID          | Text         | Unique identifier for each bus route                                                | PK  | BusRoute               |
| BusRouteName        | Text         | Name of the bus route                                                               | -   | BusRoute               |
| DateID              | Date         | Unique identified used to link records to the Calendar table                        | PK  | Calendar Dim           |
| MonthName           | Text         | Full name of the month                                                              | -   | Calendar Dim           |
| MonthNumber         | Whole Number | Numeric representation of the month (1-12)                                          | -   | Calendar Dim           |
| MonthShortForm      | Text         | 3 letter abbreviation of the month, used for concise reporting & visualizations     | -   | Calendar Dim           |
| Quarter             | Whole Number | Quarter of the year (Q1-Q4)                                                         | -   | Calendar Dim           |
| Weekday             | Text         | Name of the day of the week                                                         | -   | Calendar Dim           |
| WeekdayShortForm    | Text         | Abbrevated name of the day of the week, used for concise reporting & visualizations | -   | Calendar Dim           |
| WeekNumber          | Whole Number | Week number of the year                                                             | -   | Calendar Dim           |
| Year                | Whole Number | Calendar year of the date                                                           | -   | Calendar Dim           |
| AlertID             | Whole Number | Unique identified for each alert record                                             | PK  | Historical Alerts Fact |
| AlertSeverityRange  | Text         | Severity level associated with the alert eg 0, 1-19, 20-39                          | -   | Historical Alerts Fact |
| BusRouteID          | Text         | Links each alert to the BusRoute table                                              | FK  | Historical Alerts Fact |
| DateID              | Date         | Links each alert to the Calendar table                                              | FK  | Historical Alerts Fact |
| HourID              | Whole Number | Links each alert to the Time table                                                  | FK  | Historical Alerts Fact |
| RouteStatus         | Text         | Status of the route at the time of the alert                                        | -   | Historical Alerts Fact |
| RouteStatusCategory | Text         | Category grouping assigned to the route status                                      | -   | Historical Alerts Fact |
| SystemwideRouteID   | Text         | Links each alert to the SystemwideRoute table                                       | FK  | Historical Alerts Fact |